FAKULTÄT FÜR MATHEMATIK UND INFORMATIK
UNIVERSITÄT LEIPZIG.

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# Lina María Simbaqueba M.

## Education.

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| 2025- | PhD student, Universität Leipzig. |
| 2023-2025 | **Ms, Mathematics,** University of Victoria. |
| 2018-2022 | **Bs, Mathematics**, Universidad Nacional de Colombia. |

## Employment.

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| 2023-2025 | **Teaching Assistant,** University of Victoria. |
| 2022-2022 | **Research Assistant,** Universidad Nacional de Colombia. |
| 2022-2022 | **Research Assistant,** University of Victoria – Mitacs Globalink. |
| 2021-2021 | **Teaching Assistant,** Universidad Nacional de Colombia. |

## Publications and Preprints.

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| 2024 | J. Noel, A. Ranganathan, L.M. Simbaqueba, Forcing Quasirandomness in a Regular Tournament, Accepted in Innovations in Graph Theory. |
| 2024 | J.-L. Baril, D. Colmenares, J. Ramirez, D. Silva, L.M. Simbaqueba, D. Toquica, Consecutive pattern-avoidance in Catalan words according to the last symbol, RAIRO-Theoretical Informatics and Applications, 58 (2024). |
| 2024 | R. Flórez, J. González, M. Matijasevick, C. Pardo, L.M. Simbaqueba, and F. Velandia, Lattice paths in corridors and cyclic corridors, Contributions to discrete mathematics, 19 (2), (2024), 35-55. |
| 2023 | B. Natalie, C. Gabriel, J. Noel, L. M. Simbaqueba, Sidorenko-Type Inequalities for Pairs of Trees, Random Structures Algorithms. 67 (1) (2025) Paper e70026. |
| 2022 | J.-L. Baril, J.-L. Ramirez, L.-M. Simbaqueba, Equivalence classes of skew Dyck paths Modulo some patterns, Integers: Electronic Journal of Combinatorial Number Theory, 22 (2022). |
| 2021 | J.-L. Baril, J.-L. Ramirez, L.-M. Simbaqueba, Counting prefixes of skew Dyck paths, Journal of Integer Sequences, 2021, Vol. 24, Article 21.8.2. |

## Talks.

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| 2020 | VI Regional Conference of Number Theory, Universidad del Cauca. |
| 2020 | Meeting of Math Students, Universidad de Antioquia. |
| 2022 | Seminario Sabanero de Combinatoria, Universidad Javeriana. |
| 2022 | XI Simposio Nororiental de Matemáticas, Universidad Industrial de Santander. |
| 2024 | Discrete Mathematics Days, DMD 2024, Alcalá de Henares. |
| 2024 | Summit280, Budapest. |
| 2025 | CanaDAM 2025, Ottawa. |

## Teaching and Tutoring.

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| 2022-2023 | **Círculos Matemáticos**, Sociedad Colombiana de Matemáticas. This program seeks to promote interest in mathematics to high school students. |
| 2020-2021 | **Volunteer in TRAUM.** This program promotes equality through accessible education for Colombian teenagers. |
| 2021-2022 | First-year **mathematics courses**, Universidad Nacional de Colombia. |

## Honors and Awards.

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| 2019 | Undergraduate degree payment exemption. |
| 2023 | Best Undergraduate Thesis Award, Universidad Nacional de Colombia. |
| 2025 | First place in natural sciences, Concurso Nacional Otto de Greiff. |

## Courses at higher level.

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| 2023 | **Algebraic methods in combinatorics**: an introduction to the use of linear algebra tools such as the polynomial method to solve problems in extremal combinatorics. The duration was 3 hours every week for 4 months. University of Victoria. |
| 2024 | **Structural graph theory**: an overview of the main properties studied in graphs such as Hamiltonian cycles, chromatic number, independence number into different families of graphs such as permutation graphs, chordal graphs, and interval graphs. Duration: 3 hours every week for 4 months. University of Victoria. |
| 2024 | **Extremal combinatorics**: an introduction to some classical problems of extremal combinatorics such as intersecting set theory, Ramsey theory, Turán theory, and regularity lemma. University of Victoria. |
| 2025 | **Convex polytopes**: an introduction to the theory of convex polytopes. Duration: 8 hours. Max Planck Institute for Mathematics in the Sciences. |

## Extracurricular Courses.

- Combinatorics I by Dr. Robert Morris at IMPA Brasil.
- Algorithmic Toolbox by the University of California, San Diego on the Coursera platform.

## References.


- **José Luis Ramírez** (supervisor), jlramirezr@unal.edu.co, Universidad Nacional de Colombia.
- **Jonathan Noel** (supervisor), noelj@uvic.ca, University of Victoria.